EA Credit: Enhanced Refrigerant Management

This credit applies to

- BD+C: New Construction (1 point)
- BD+C: Core & Shell (1 point)
- BD+C: Schools (1 point)
- BD+C: Retail (1 point)
- BD+C: Data Centers (1 point)
- BD+C: Warehouses & Distribution Centers (1 point)
- BD+C: Hospitality (1 point)
- BD+C: Healthcare (1 point)

INTENT

To eliminate ozone depletion and global warming potential, and support early compliance with the Montreal Protocol, including the Kigali Amendment, while minimizing direct contributions to climate change.

REQUIREMENTS

NC, CS, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Option 1. No Refrigerants or Low-Impact Refrigerants (1 point)
Do not use refrigerants, or use only refrigerants (naturally occurring or synthetic) that have an ozone depletion potential (ODP) of zero and a global warming potential (GWP) of less than 50.

OR

Option 2. Calculation of Refrigerant Impact (1 point)
Comply with ASHRAE Standard 15-2019: Safety Standard for Refrigeration Systems, or USGBC-approved equivalent, as applicable to the project scope.

Develop and implement a refrigerant management plan that addresses leak detection, system retrofit, and end of life disposal for all HVAC&R systems containing more than 0.5 pound (225 grams) of refrigerant.

Select refrigerants that are used in heating, ventilating, air-conditioning, and refrigeration (HVAC&R) equipment to minimize or eliminate the emission of compounds that contribute to ozone depletion and climate change. The combination of all new and existing base building and tenant HVAC&R equipment that serve the project must comply with the following formula:

IP units	SI units
$LCGWP + LCODP \times 10^5 \leq 100$	$LCGWP + LCODP \times 10^5 \leq 13$
Calculation definitions for LCGWP + LCODP x $10^5 \leq 100$ (IP units)	**Calculation definitions for LCGWP + LCODP x $10^5 \leq 13$ (SI units)**
LCODP = [ODPr x (Lr x Life +Mr) x Rc]/Life	LCODP = [ODPr x (Lr x Life +Mr) x Rc]/Life
LCGWP = [GWPr x (Lr x Life +Mr) x Rc]/Life	LCGWP = [GWPr x (Lr x Life +Mr) x Rc]/Life
LCODP: Lifecycle Ozone Depletion Potential (lb CFC 11/Ton-Year)	LCODP: Lifecycle Ozone Depletion Potential (kg CFC 11/(kW/year))

LCGWP: Lifecycle Direct Global Warming Potential ($lb\ CO_2$/Ton-Year)	LCGWP: Lifecycle Direct Global Warming Potential ($kg\ CO_2$/kW-year)
GWPr: Global Warming Potential of Refrigerant (0 to 12,000 $lb\ CO_2$/lbr)	GWPr: Global Warming Potential of Refrigerant (0 to 12,000 $kg\ CO_2$/kg r)
ODPr: Ozone Depletion Potential of Refrigerant (0 to 0.2 lb CFC 11/lbr)	ODPr: Ozone Depletion Potential of Refrigerant (0 to 0.2 kg CFC 11/kg r)
Lr: Refrigerant Leakage Rate (2.0%)	Lr: Refrigerant Leakage Rate (2.0%)
Mr: End-of-life Refrigerant Loss (10%)	Mr: End-of-life Refrigerant Loss (10%)
Rc: Refrigerant Charge (0.5 to 5.0 lbs. of refrigerant per ton of gross AHRI rated cooling capacity)	Rc: Refrigerant Charge (0.065 to 0.65 kg of refrigerant per kW of AHRI rated or Eurovent Certified cooling capacity)
Life: Equipment Life (10 years; default based on equipment type, unless otherwise demonstrated)	Life: Equipment Life (10 years; default based on equipment type, unless otherwise demonstrated)

For multiple types of equipment, calculate a weighted average of all base building HVAC&R equipment, using the following formula:

IP units	SI units
$$\frac{\left[\sum \left(LCGWP + LCODP \times 10^5 \right) \times Qunit \right]}{Qtotal} \leq 100$$	$$\frac{\left[\sum \left(LCGWP + LCODP \times 10^5 \right) \times Qunit \right]}{Qtotal} \leq 13$$

Calculation definitions for $[\sum (LCGWP + LCODP \times 10^5) \times Qunit] / Qtotal \leq 100$ (IP units)	Calculation definitions for $[\sum (LCGWP + LCODP \times 10^5) \times Qunit] / Qtotal \leq 13$ (SI units)
Qunit = Gross AHRI rated cooling capacity of an individual HVAC or refrigeration unit (Tons)	Qunit = Eurovent Certified cooling capacity of an individual HVAC or refrigeration unit (kW)
Qtotal = Total gross AHRI rated cooling capacity of all HVAC or refrigeration	Qtotal = Total Eurovent Certified cooling capacity of all HVAC or refrigeration (kW)

Retail NC

Meet Option 1 or 2 for all HVAC systems.

Stores with commercial refrigeration systems must comply with the following.

- Use only non-ozone-depleting refrigerants.
- Select equipment with an average HFC refrigerant charge of no more than 1.75 pounds of refrigerant per 1,000 Btu/h (2.72 kg of refrigerant per kW) total evaporator cooling load.

- Demonstrate a predicted store-wide annual refrigerant emissions rate of no more than 15%. Conduct leak testing using the procedures in GreenChill's best practices guideline for leak tightness at installation.

Alternatively, stores with commercial refrigeration systems may provide proof of attainment of EPA GreenChill's silver-level store certification for newly constructed stores.

GUIDANCE

Refer to the LEED v4 reference guide, with the following edits and additions:

Behind the Intent

While the Montreal Protocol provided for the phase-out of CFC and HCFC refrigerants due to their ozone depletion potential, it led the industry shift toward the usage of hydrofluorocarbons (HFCs), which are potent greenhouse gases. In the Kigali Amendment to the Montreal Protocol, adopted in 2016, 197 countries committed to cut the production and consumption of HFCs by more than 80 percent over the next 30 years.

Alternatives to HFCs have been developed to provide lower-GWP refrigerant options, although the availability of low-GWP refrigerants varies between applications. The transition to lower-GWP refrigerant options requires an increased focus on safety, as many "next generation" refrigerants can pose flammability and/or toxicity risks if not properly managed. ASHRAE Standard 15-2019: Safety Standard for Refrigeration Systems provides essential guidance to manufacturers, design engineers and operators who need to stay current with new air conditioning and refrigerating requirements. The selection of refrigerants and their operating systems should be based on a holistic analysis of multiple criteria, including safety, environmental impacts, energy efficiency and cost.

In addition to protecting human health, effective refrigerant management during processes of design, manufacturing, operation, systems servicing, and end of life helps to reduce global emissions. The majority of refrigerant emissions happen at end of life, so the effective disposal of refrigerants currently in circulation (including CFCs, HCFCs and HFCs) is essential. A robust refrigerant management plan provides a framework for building owners and operators to properly manage these chemicals and minimize direct contributions to climate change.

Step-by-Step

Step 1. Document compliance with ASHRAE Standard 15-2019

Review ASHRAE Standard 15-2019: Safety Standard for Refrigeration Systems. The standard establishes procedures for the safe design, construction, installation and operation of refrigerant systems. Note that Standard 15 must be used with ASHRAE Standard 34-2019 Designation and Safety Classification of Refrigerants, which describes a shorthand way of naming refrigerants and assigns safety classifications based on toxicity and flammability data.

Ensure that all refrigeration equipment installed in the project comply with ASHRAE Standard 15-2019 requirements for restrictions on refrigerant use (section 7), installation restrictions (section 8), design and construction of equipment and systems (section 9), operation and testing (section 10) and general requirements (section 11), as applicable to the project scope.

Step 2. Develop and implement refrigerant management plan

Develop and implement a refrigerant management plan. The plan must include a complete inventory of all refrigerants used in the project and address refrigerant leak detection, system retrofit, and proper end of life disposal for all HVAC&R systems containing more than 0.5 pound (225 gram) of refrigerant.

The plan must identify the individuals responsible for the care and maintenance of refrigerant systems, regular leak detection and monitoring, record-keeping and training of personnel necessary to execute the plan.

All contractors or service technicians performing work on any HVAC&R systems shall have access to the refrigerant management plan and, upon completion of the work, provide documentation that confirms continued compliance with the plan.

Step 3. Calculate refrigerant impact of proposed systems

Refer to the LEED v4 reference guide.

Step 4. Incorporate design criteria into project plans and specifications

Refer to the LEED v4 reference guide.

Further Explanation

Referenced Standards

- ASHRAE Standard 15-2019: Safety Standard for Refrigeration Systems

Required Documentation

Refer to the LEED v4 Reference Guide content, plus the following additions for Option 2:

- Confirmation that project HVAC&R systems comply with ASHRAE Standard 15
- Refrigerant management plan

Connection to Ongoing Building Performance

- LEED O+M EA credit Enhanced Refrigerant Management: Effective refrigerant selection and management, especially at the point of disposal, is a critical strategy for addressing climate change and minimizing the release of building-related emissions into the atmosphere. Additionally, careful consideration of refrigerants used in HVAC&R equipment can improve performance and reduce operating costs throughout the building life cycle.